Precambrian Research 292 (2017) 1–12

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Fennoscandia before Nuna/Columbia: Paleomagnetism of 1.98–1.96 Ga mafic rocks of the Karelian craton and paleogeographic implications

N.V. Lubnina [a,*], S.A. Pisarevsky [b,g], A.V. Stepanova [c,d], S.V. Bogdanova [e,f], S.J. Sokolov [c]

[a] Faculty of Geology, M.V. Lomonosov Moscow State University, Leninskiye Gory, 1, Moscow 119991, Russia
[b] Earth Dynamics Research Group, ARC Centre of Excellence for Core to Crust Fluid Systems (CCFS) and The Institute for Geoscience Research (TIGeR), Department of Applied Geology, Curtin University, GPO Box U1987, WA 6845, Australia
[c] Institute of Geology, Karelian Research Center, Pushkinskaya st., 11, Petrozavodsk 185910, Russia
[d] Institute of Geology of Ore Deposits, Petrography, Mineralogy and Geochemistry RAS, Staromonetny, 35, Moscow 119017, Russia
[e] Department of Geology, Lund University, Sölvegatan 12, SE 223 62 Lund, Sweden
[f] Kazan Federal University, 18 Kremlyovskaya str., Kazan 420008, Russia
[g] Institute of the Earth's Crust, Siberian Branch of the Russian Academy of Sciences, Irkutsk, 128 Lermontov Street, 664033, Russia

ARTICLE INFO

Article history:
Received 10 September 2016
Revised 9 January 2017
Accepted 18 January 2017
Available online 21 January 2017

Keywords:
Paleoproterozoic mafic dykes and sills
Baltica
Karelia
Paleomagnetism
Key pole
Nuna/Columbia supercontinent

ABSTRACT

Numerous mafic dykes, sills and intrusions with ages between 1985 Ma and 1960 Ma are exposed near the Onega Lake in southern Karelia, Russia. The paleomagnetic analysis of these rocks has revealed a stable remanence with directions belonging to two groups. The directions of the first group characterize ten intrusions including the dated 1970 ± 3 Ma Unoi sill and 1976 ± 9 Ma Suna River Canyon dolerite, the corresponding paleomagnetic pole is 44.4°N, 101.5°E, A_{95} = 6.3°. The second group comprises two intrusions including the 1984 ± 8 Ma Pudozhgora intrusion and Krestoviy Navolok dyke with the corresponding paleopole calculated from 5 site mean poles is 60.9°N, 144.8°E, A_{95} = 6.8°. Both remanence directions are supported by robust baked contact tests. We propose the first group's pole as the key 1975 Ma Fennoscandian pole. The second one is well dated, but based only on two intrusions without proper averaging of the paleosecular variations. We have also carried out a complimentary paleomagnetic study of the previously investigated 2504 Ma Shalskiy gabbronorite dyke. The remanence of this dyke is now supported by the inverse contact test and statistics can be improved. Using our 1975 Ma pole together with coeval poles from Superior, Slave and Amazonia cratons we propose a provisional 1975 Ma paleogeographic reconstruction.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The 2.0–1.8 Ga time interval in Earth's history is marked by world-spread orogenic events (e.g. Zhao et al., 2009 and references therein). These events initiated assemblies of Laurentia (the cratonic part of North America), Baltica, Siberia, Australia, Kalahari cratons (e.g. Hoffman, 1988; Bogdanova et al., 2008; Rosen et al., 1994; Pisarevsky et al., 2008; Cawood and Korsch, 2008; Hanson et al., 2011) and accretionary growth of others (e.g. Brito Neves, 2011). Most of these and other cratons eventually collided to form the Paleo-Mesoproterozoic supercontinent Nuna (or Columbia) (Rogers and Santosh, 2002; Zhao et al., 2004, 2006; Meert, 2012 and references therein). The time of the final assembly of Nuna/Columbia is debated and estimated between ca. 1800 Ma (Zhang et al., 2012 and references therein) and ca. 1600 Ma (Pisarevsky et al., 2014 and references therein).

Recent studies demonstrate that Baltica (the East European Craton) has not been assembled until ca. 1.8–1.7 Ga (Bogdanova et al., 2008; 2013; Lubnina et al., 2015a). At ca 2.0 Ga, at least three protocratons – Kola, Karelia and Volgo-Sarmatia were separated by oceans (Melezhik and Hanski, 2013; Bogdanova et al., 2008, 2013). The small Kola Ocean between Kola and Karelia was closed at ca. 1.94 Ga (Daly et al., 2006; Melezhik and Hanski, 2013). Starting from ca. 1.90 Ga, the united Kola-Karelia grew in SW- to SSW direction (hereafter – in present day coordinates) by consequent accretion of island arcs and microcontinents to amalgamate Fennoscandia (Bogdanova et al., 2015).

Several early Paleoproterozoic paleogeographic reconstructions around 2.5–2.4 Ga (e.g. Bleeker and Ernst, 2006; Söderlund et al., 2010; Salminen et al., 2014; Pisarevsky et al., 2015) and late Paleoproterozoic, younger than 1.8 Ga (e.g. Pisarevsky and Bylund, 2010; Zhang et al., 2012; Pisarevsky et al., 2014) have been

* Corresponding author.
E-mail address: natalia.lubnina@gmail.com (N.V. Lubnina).

http://dx.doi.org/10.1016/j.precamres.2017.01.011
0301-9268/© 2017 Elsevier B.V. All rights reserved.

published in recent years. Mutual positions of the Baltica and Laurentia cratons are extremely different for these time slices. These differential motions imply major plate reorganization during the mid-Paleoproterozoic (~2.0 Ga). The details of these movements are unclear because of a paucity of reliable paleomagnetic data. Buchan et al. (1998) reported a reliable 1998 ± 2 Ma paleomagnetic pole for the Minto dykes (Superior craton), but no reliable Karelian poles for Baltica with similar ages have been published so far. Pisarevsky and Sokolov (1999) reported a virtual geomagnetic pole (VGP) from the 1975 ± 24 Ma Konchozero sill in southern Karelia, Russia (Sm–Nd, Puchtel et al., 1998). However, this pole represents just one intrusion and the geomagnetic secular variations probably have not been averaged. Recent studies in southern Karelia (Lubnina et al., 2015b, 2016 and references therein) have demonstrated that the 1.98–1.97 Ga mafic magmatism was widespread in this area and is represented by numerous dykes and sills (Fig. 1). Pasenko and Lubnina (2014) have reported that many of these intrusions carry a stable remanence. In this paper, we present the results of paleomagnetic study of these intrusions.

The c. 1.99–1.96 Ga reliable paleomagnetic poles from Slave and Amazonia proto-continents have been also published (Evans and Hoye, 1981; Bispo-Santos et al., 2014; Buchan et al., 2016). Such concentration of reliable coeval paleopoles from different continents is very rare for Paleoproterozoic, so our new data provide an opportunity to reconstruct paleo-latitudinal positions and orientation of five proto-cratons at 1.99–1.96 Ga and to test the postulated by some workers hypothesis of the supercontinental assembly at c. 2.1–1.8 Ga (e.g. Zhao et al., 2009).

2. Geology and sampling

The Archean Karelian Craton was formed through the amalgamation of various terranes at ca. 2.70 Ma (Slabunov et al., 2006). The oldest Vodlozero terrane (Fig. 1) is located in its southeastern part. There, 3.30–2.95 Ga granite-greenstone rocks compose a basement of the Onega Paleoproterozoic structure (Slabunov et al., 2006; Glushanin et al., 2011), or the Onega Basin (Melezhik et al., 2013) (Fig. 1). The Onega Basin is filled by Paleoproterozoic supracrustal rocks of Sumian (2.50–2.40 Ga), Sariolian (2.40–2.30 Ga), Jatulian (2.30–2.10 Ga), Ludicovian (2.10–1.92 Ga), Kalevian (1.92–1.80 Ga), and Vepsian (1.80–1.65 Ga) Formations, with a cumulative thickness of up to 10 km (Glushanin et al., 2011 and references therein).

At least two major pulses of mafic magmatism have been recognized in this area: at 2.50–2.45 Ga (Amelin et al., 1995; Puchtel et al., 1997; Kulikov et al., 2010) and at ca. 1.98–1.96 Ga (Puchtel et al., 1998; Philippov et al., 2007; Stepanova et al., 2014). The older event is characterized by komatiitic, basaltic, and felsic volcanic rocks and siliciclastic sediments formed in rift-related volcano-sedimentary belts synchronous with layered mafic intrusions, granite massifs, and gabbronorite dyke swarms

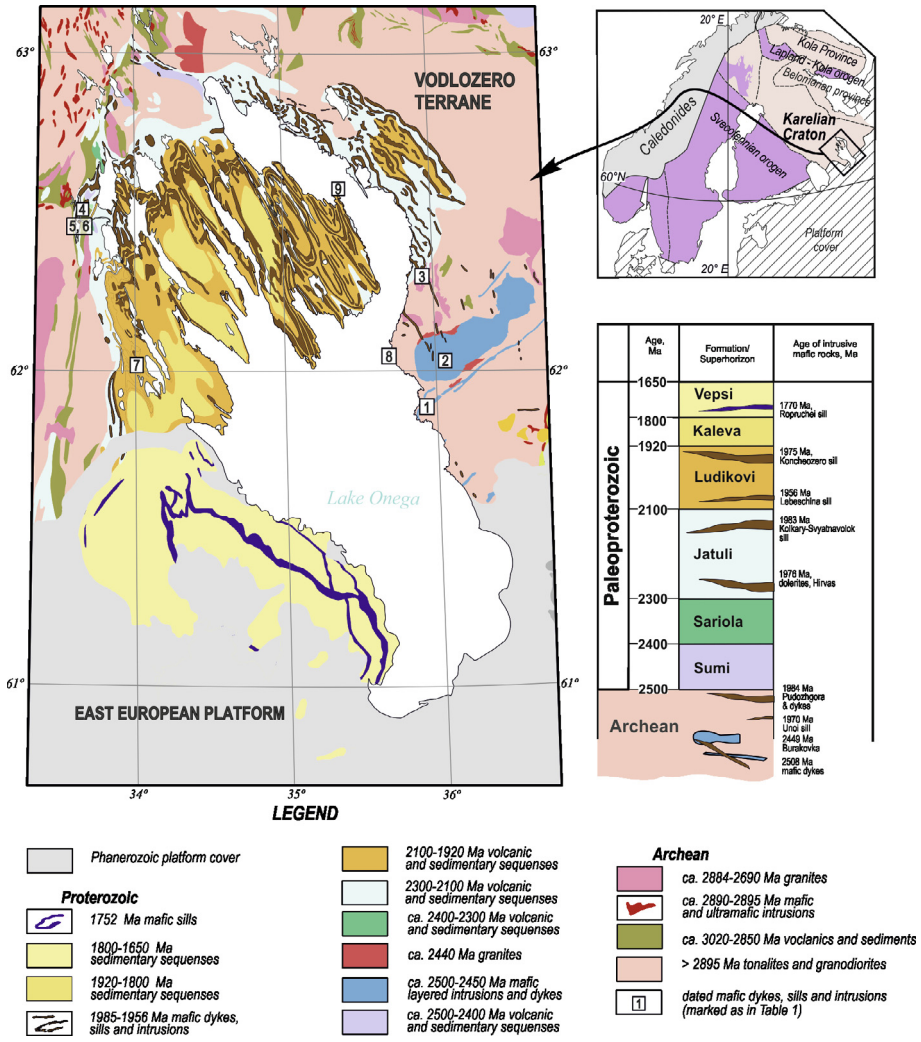


Fig. 1. Geological map and sample localities (numbers are as in Table 2) of the Onega Lake area. After Kulikov et al. (2016).

Table 1
Ages of mafic dykes, sills and intrusions from the Onega Basin and adjacent area.

№ on Fig. 1	Site	Rock	Mineral, rock	Method	Age, Ma	Reference
1	Shalskiy and Avdeev dykes	Gabbronorite	Baddeleyite	U–Pb TIMS	ca 2504	Bleeker et al. (2008)
2	Burakovka intrusion	Gabbronorite	Zircon	U–Pb TIMS	2449 ± 1	Amelin et al. (1995)
3	Pudozhgora intrusion	Gabbrodolerite	Zircon	U–Pb SIMS (SHRIMP)	1984 ± 8	Philippov et al. (2007)
4	Koikary-Svyatnavolok mafic sill	Dolrite	Zircon	U–Pb SIMS (SHRIMP)	1983 ± 7	Philippov et al. (2007)
5	Suna River Canyon	Dolerite	Zircon	U–Pb SIMS (SHRIMP)	1976 ± 9	Kulikov et al. (1999)
6	Suna River Canyon	Dolerite	Zircon baddeleyite	CA ID TIMS	1975.3 ± 3	Martin et al. (2015)
7	Konchozero sill	Dolerite	Whole rock	Sm–Nd	1975 ± 24	Puchtel et al. (1998)
8	Unoi sill	Gabbrodolerite	Baddeleyite	U–Pb TIMS	1970 ± 3	Lubnina et al. (2016)
9	Zaonega Peninsula	Dolerite	Zircon	U–Pb SIMS (SHRIMP)	1956 ± 6	Stepanova et al. (2014)

(Stepanova et al., 2015 and references therein). It is generally assumed that melting of the mantle and crust at ca. 2.50–2.45 Ga was caused by the ascent of a mantle plume. The younger event is manifested by diverse, probably mantle plume-related mafic complexes that include picrites, high- and low-Ti tholeiites, subalkaline basalts, syenites, carbonatites, and kimberlites (Puchtel et al., 1998; Stepanova et al., 2015; Ernst, 2014 and references therein).

Although many Paleoproterozoic mafic dykes and sills are exposed in the Onega Basin and in the adjacent area, only few of them have been isotopically dated (Table 1).

These geochronological data together with petrological and geochemical studies (Lubnina et al., 2016) suggest a tentative subdivision of the 1.98–1.95 Ga magmatism into three pulses close one to each other in composition and geochemical characteristics (Supplementary data I and II).

1. *Ca. 1985 Ma pulse*, represented by the Koikary-Svyatnavolok and Pudozhgora sills of the Onega Basin and in the adjacent areas together with a series of sills and NNW-trending dykes to the north of our study area (Golubev and Svetov, 1983; Trofimov and Golubev, 2008). These are mostly large bodies with up to 200 m thickness, which are traced up to 25 km (Glushanin et al., 2011). They are differentiated intrusions characterized by a presence of ore horizons enriched with titanomagnetite. Another common feature of these bodies is the presence of diorites in their roofs with high (>6 wt.%) content of Na_2O and high contents of incompatible elements (Supplementary data I and II). The composition of the ca. 1985 Ma mafic rocks corresponds to high-Fe tholeiitic basalts.
2. *Ca. 1975 Ma pulse*, characterized by the high-Mg Konchozero sill and some coeval basalts, high-Fe Suna River Canyon dolerites and the high-Fe Unoi sill (Supplementary data I and II).
3. *Ca. 1960 Ma pulse*, defined by one 1956 ± 6 Ma dolerite sill in the central part of the Onega Basin (Stepanova et al., 2014) and the ca. 1962 Ma dolerite dykes in its north-western part (Martin et al., 2015). Also, many geochemically similar sills are reported from the Upper Ludicovian sequences in central and western parts of the basin (Kulikov and Kulikova, 2008; Stepanova et al., 2014). In general, they are geochemically similar to the dolerites of two older pulses (Supplementary data I and II).

Table 2
Paleomagnetic directions and poles of mafic intrusions in the Onega Lake area, Karelia, Russia.

#	Intrusion	Age (Ma)	Slat (°N)	Slong (°E)	B/N	D(°)	I(°)	k	α_{95}(°)	Plat (°N)	Plong (°)	Dp(°)	Dm (°)
1	Pudozhgora intrusion	1984 ± 8	62.3013	35.9038	3/64	38.0	60.5	33.6	3.1	59.4	150.9	3.6	4.7
2	Gabnev sill		62.2938	35.8460	3/19	60.8	73.1	32.7	6.0	60.8	104.4	9.4	10.6
3	Gabnev dyke		62.2938	35.8460	1/15	97.3	70.5	52.3	5.3	43.5	88.1	8.0	9.2
4	Unoi sill	1970 ± 3	61.9875	35.6948	5/89	75.3	65.6	18.6	3.6	47.2	108.7	4.7	5.8
5	Krestoviy Navolok dyke*		61.8422	35.8521	2/30	42.0	65.3	37.8	4.3	62.4	138.0	5.7	7.0
6	Krestoviy Navolok, host granite		61.8422	35.8521	2/9	153.2	32.5	18.1	12.4	7.1	241.5	7.9	14.0
7	Deda young dyke*		61.8187	35.8598	2/40	83.0	60.4	21.2	5.0	38.7	108.6	5.8	7.6
8	Deda old dyke	ca 2504	61.8187	35.8598	1/24	173.0	-5.7	23.2	6.3	30.8	224.0	3.2	6.3
9	Deda host granite		61.8187	35.8598	1/9	245.9	-64.8	36.4	8.7	50.7	117.1	11.2	13.9
10	Gorely dyke 1*		61.8197	35.8797	2/9	75.6	64.2	45.7	7.7	45.7	110.3	9.8	12.3
11	Gorely dyke 2*		61.8197	35.8797	2/10	80.8	61.1	36.9	8.1	40.4	109.6	9.5	12.4
12	Gorely host granite		61.8197	35.8797	1/15	236.6	-65.5	168.1	3.0	55.5	123.9	3.9	4.8
13	Shala dyke 1		61.8237	35.8802	1/10	89.8	60.1	23.1	10.3	35.5	103.4	11.8	15.6
14	Shala dyke 2		61.8237	35.8802	1/11	89.5	63.5	20.4	10.4	38.9	100.9	13.0	16.4
15	Shala dyke 3		61.8237	35.8802	1/9	76.8	73.3	50.9	7.3	54.3	95.0	11.7	13.0
16	Shala host granite		61.8237	35.8802	1/11	263.4	-70.3	27.4	8.9	48.4	96.6	13.2	15.3
17	Suna Canyon dolerite (overpr.)		62.4568	33.6692	5/17	20.7	46.3	75.4	4.1	52.6	182.6	3.4	5.3
18	Suna Canyon dolerite	1976 ± 9	62.4568	33.6692	6/19	103.9	66.1	30.3	6.2	36.1	86.5	8.3	10.2
	Mean 2–4, 7, 10–11, 13–15, 18 (1975 Ma)				10 intrusions					44.4	101.5	A_{95} = 6.3	
	Mean 1, 5 (1985 Ma)				5 sites from 2 intrusions					60.9	144.8	A_{95} = 6.8	
	Mean 1–5, 7, 10–11, 13–15, 18 (both 1975 Ma and 1985 Ma)				12 intrusions					48.0	106.4	A_{95} = 8.2	
	Mean 9, 12, 16 (2630 ± 10 Ma host granite)				3 sites					61.8	117.0	A_{95} = 24.2	

Slat/Slong = location coordinates; B/N = sites/samples; D/I = declination/inclination; k = Fisher's precision parameter; α_{95} = the semi-angle of the 95% cone of confidence; Plat, Plong = latitude, longitude of the paleopole; Dp, Dm = the semi-axes of the cone of confidence about the pole at the 95% probability level.
* Baked host samples included.

However, we should note that available ages for the first and second pulses are overlapping, so it is possible that they belong to just one pulse.

The 1985–1960 Ma mafic dykes and sills are geochemically different from the 2504 Ma high-Mg Avdeev and Shalskiy dykes in the same areas. Until recently only the 2450 Ma Burakovka intrusion, the 2504 Ma Avdeev and Shalskiy dykes and the 1975 Ma Konchozero sill (Table 1) have been studied paleomagnetically (Mertanen et al., 2006 and references therein; Pisarevsky and Sokolov, 1999).

Pasenko and Lubnina (2014) have reported preliminary paleomagnetic results from the ca. 1985 Ma Pudozhgora intrusion and 1970 Ma Unoi sill (Table 1), as well as from two previously undated intrusions in the eastern part of the Onega basin (Table 2). Here, we present the results of the final paleomagnetic analysis of these intrusions, and also of several other dykes in the Onega basin and of the ca. 1975 Ma Suna River Canyon sill (Table 1). We collected c. 500 oriented cores and block samples from 12 mafic intrusions and host rocks (Figs. 1 and 6). Details about geochemistry, geochronology and magnetic fabric of most of these intrusions have been recently published (Lubnina et al., 2016).

3. Petrography and rock magnetism

All studied rocks are dolerites with typical sub-ophitic texture and composed mainly of clinopyroxene (augite) and plagioclase, with only minor amounts of other minerals like Fe–Ti-oxides, quartz, and scarce olivine. The dolerites vary in grain size and modal proportions depending on dyke or sill thickness as well as the position within the dyke or sill. They are fresh and only weakly altered not exceeding greenschists facies. More detailed petrography is described in Lubnina et al. (2016).

The thermomagnetic analysis (changes of magnetic susceptibility with temperature) has been carried out using KLY-4S (AGICO). This analysis clearly demonstrates that low-Ti titanomagnetite is a dominant magnetic mineral in all studied mafic rocks (Fig. 2). Well-defined Hopkinson peaks (Fig. 2), indicate that the remanence is carried mostly by highly paleomagnetically stable single-domain (SD) particles. Further thermal and alternative field (AF) demagnetisations support this conclusion (see next section).

4. Paleomagnetic analysis

Remanence composition was determined by detailed stepwise AF demagnetization (⩽15 steps, up to 100 mT) using LDA−3A−AF Demagnetizer and the JR−6 spinner-magnetometer (AGICO). Stepwise thermal demagnetization (⩽20 steps, to 595 °C), using the TD48 (ASC Scientific) was also applied. To monitor possible mineralogical changes during heating, magnetic susceptibility was measured in selected samples after each heating step using KLY-4S Kappa-bridge. Magnetisation vectors were isolated using Principal Component Analysis (Kirschvink, 1980). All vectors were defined with a minimum of four data points and a maximum angular deviation (MAD) of 5°.

In most cases one stable Characteristic Remanent Magnetization (ChRM) have been isolated by demagnetizations due to either heating at 550–600 °C or after 30–100 mT of the AF treatment (Fig. 3A–C). In few cases (mostly in the Suna River Canyon Sill) two components have been isolated (Fig. 3D−F).

In most intrusions, including the 1970 Ma Unoil Sill, the ChRM has E steep downward directions (Table 2; Fig. 3B–C, E). In two intrusions, the ca. 1985 Ma Pudozhgora and the Krestoviy Navolok, the ChRM has slightly different NE steep downward directions (Table 2; Fig. 3A), while most of the ca. 1975 Ma Suna River Canyon

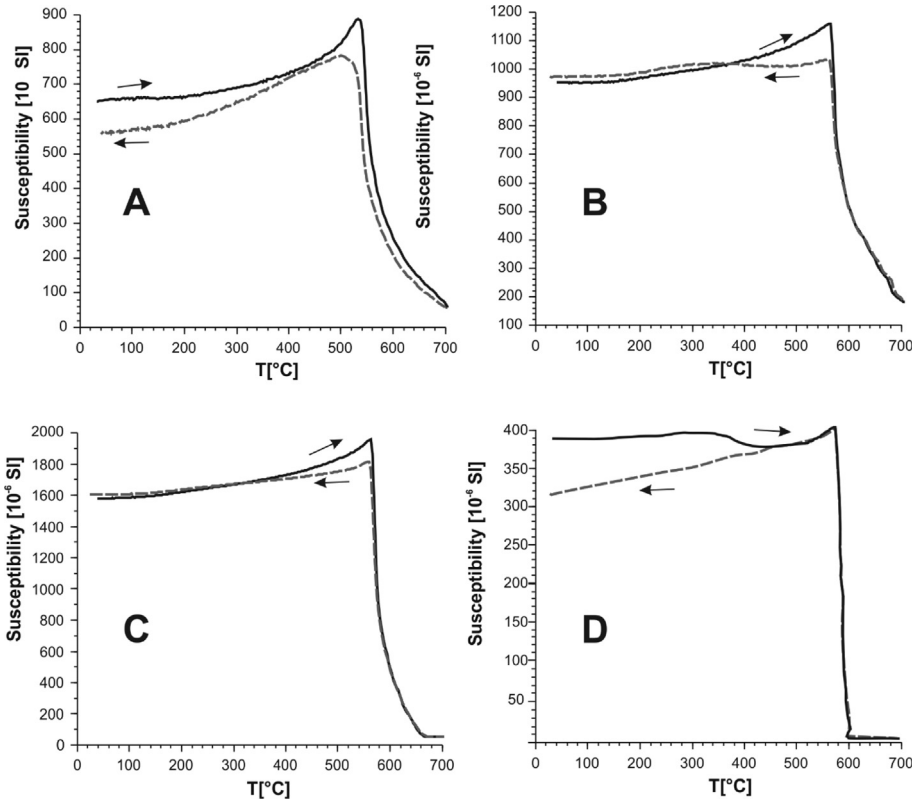


Fig. 2. The analysis of Curie points using magnetic susceptibility versus temperature curves. A – Pudozhgora intrusion; B – Krestoviy Navolok dyke; C – Shala dykes; D – Suna River Canyon dolerite.

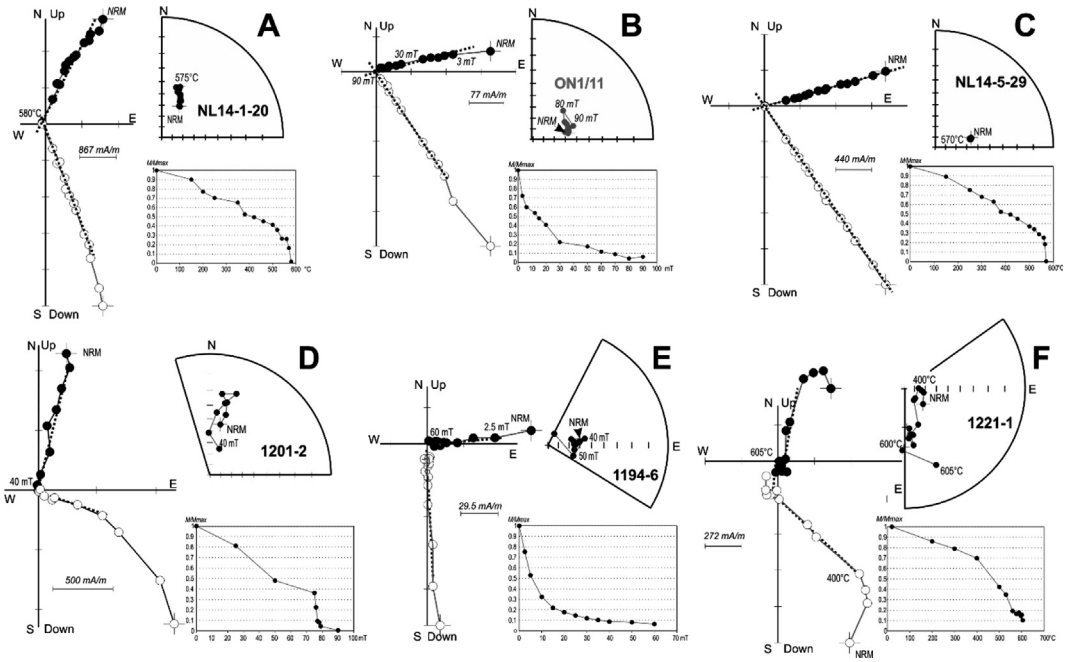


Fig. 3. Stereoplots and examples of demagnetization behaviour for studied dykes. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols. A – Krestoviy Navolok dyke; B – Deda Island dyke; C – Shala 1 dyke; D – Suna River Canyon dolerite (overprint); E – Suna River Canyon dolerite (primary ChRM); F – Suna River Canyon dolerite (two components).

dolerites carry the East downward ChRM direction (Table 2; Fig. 3E) similar to that of most studied intrusions.

However, some slightly less stable NNE ChRM directions with generally lower unblocking temperatures and coercivities has been also found in some of Suna River Canyon dolerites (Table 2, entry 17; Fig. 3D). This direction is close but distinct from the ChRM directions of the ca. 1985 Pudozhgora intrusion and of the Krestoviy Navolok dyke (Table 2, entries 1 and 5). The latter are primary in view of the positive contact test (see the next section). The age of 1976 Ma of the Suna River Canyon dolerite is significantly younger, so it is likely that the D = 20.7°, I = 46.3° direction represents an overprint (see entry 17 of Table 2). This direction is similar to the component III (D = 17°, I = 41°) of the Konchozero Sill, which was interpreted as the ca. 1760 Ma overprint (Pisarevsky and Sokolov, 1999).

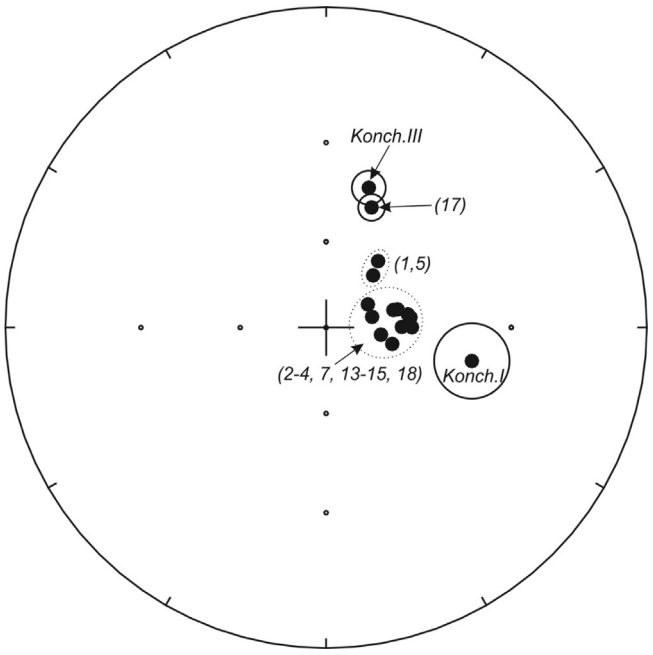


Fig. 4. Stereoplots of the mean directions of ChRM from studied intrusions and from Konchozero Sill (Pisarevsky and Sokolov, 1999). Numbers are as in Table 2.

Based on the paleomagnetic analysis we are divided the studied mafic intrusions into two groups according to their ChRM directions (Fig. 4, Table 3). The smaller group includes the 1984 Ma Pudozhgora intrusion and the undated Krestoviy Navolok dyke. The larger group consists of three sills (Gabnev, Unoi and Suna River Canyon) and seven dykes (Gabnev, Deda, Gorely 1–2, Shala 1–3). There are two dated sills of this group: the ca. 1975 Ma Suna River Canyon and the 1970 Ma Unoi sills. For simplicity, hereafter, we call these groups as the "1985 Ma intrusions" and the dykes and sills of the second group as the "1975 Ma intrusions". However, considering the error bars of available ages (Table 1) we cannot entirely dismiss a possibility that all studied intrusion belongs to one magmatic pulse. Although the combined remanence directions' distribution of the "1985 Ma intrusions" and the "1975 Ma intrusions" is elongated (Fig. 4), we calculated the mean 1985–1975 Ma pole (Table 2).

5. Contact tests

Four contacts of the Krestoviy Navolok, Deda and Gorely dykes with host granites and one, Deda dyke, with the older Shalskiy gabbronorite dyke have been sampled (Figs. 5, 6A–C). The contacts of three thin Shala dykes are not exposed, but we collected 11 samples from the host granites at the considerable distance from the dykes (Fig. 6D).

In all cases ChRM directions of the host rocks in the baked contacts of dykes are statistically undistinguishable from the ChRM directions of the studied dykes, but the ChRM directions of the unbaked host rocks are significantly different from the latter.

Table 3
2000–1960 Ma paleomagnetic poles.

#	Craton	Rockname	Age (Ma)	Plat (°N)	Plong (°E)	A_{95} (°)	Reference
1	Slave	Rifle Formation	1963 ± 3	14.0	341.0	7.7	Evans and Hoye (1981), Bowring and Grotzinger (1992)
2	W. Superior	Minto dykes	1998 ± 2	38.7	171.5	13.1	Buchan et al. (1998); recalc. by Evans and Halls (2010)
3	Karelia	Onega dykes	1970 ± 3	44.4	101.5	6.3	This study
4	Amazonia	Surumu Volcanics	1980–1960	27.4	234.8	9.8	Bispo-Santos et al. (2014)

Plat, Plong = latitude, longitude of paleopole.

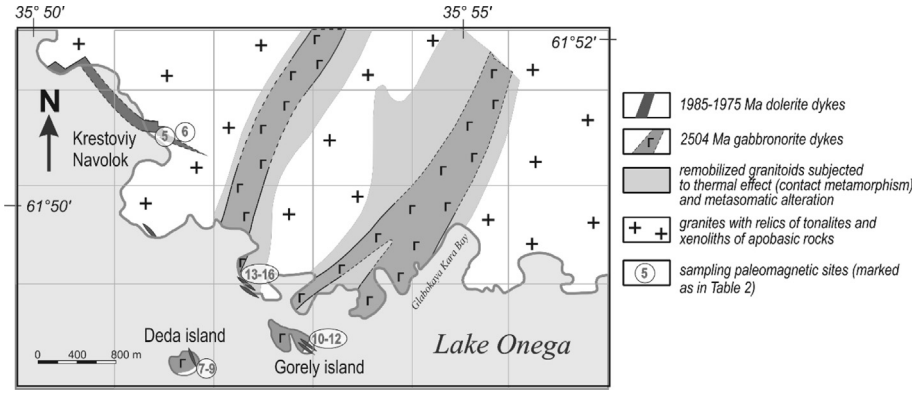


Fig. 5. Geological map of the Krestoviy Navolok – Shala area, locations of dykes 5–16 (numbers as in Table 2).

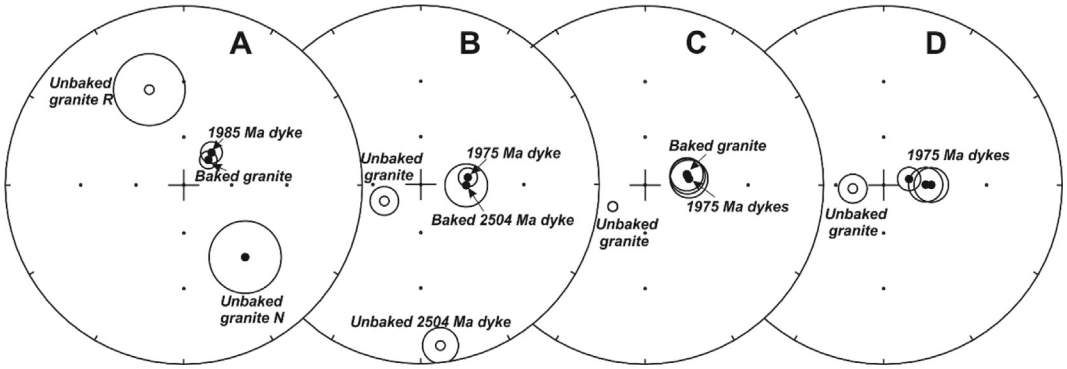


Fig. 6. Stereoplots illustrating baked contact tests: A –Krestoviy Navolok dyke; B – Deda Island dyke; C – Gorely Island dykes; D – Shala dykes.

5.1. Group of the 1985 Ma intrusions: the Krestoviy Navolok dyke

This NW–trending 60 m thick dolerite dyke is exposed in its eastern contact with Neoarchean granites (Fig. 5). The dolerites along the internal contacts are fine-grained vitreous and carry a stable remanence (see Section 4). Fig. 7 shows some details in the paleomagnetic behaviour of this dyke and host granites in the contact area. The high temperature ChRM direction is close to that of the Pudozhgora intrusion (Table 2, entries 1 and 5; Fig. 4; Fig. 7A). The mean ChRM direction of the baked contact granites collected within the first tens of centimetres from the dyke (D = 44.4°, I = 67.8°, k = 64.4, α_{95} = 5.4°) is practically undistinguishable (Fig. 6A and Fig. 7B) from the dyke ChRM direction (D = 40.6°, I = 63.6°, k = 29.5, α_{95} = 6.5°). The unbaked granite sample taken at 2 m from the dyke yields the two-component remanence (Fig. 7C). Direction of the less stable component is close to that of the dyke's ChRM. The high-temperature component is SSE with moderate downward inclination, similar to the mean ChRM direction of the granite samples collected at ca. 100 m from the dyke (D = 139.4°, I = 34.4°, k = 49.3, α_{95} = 17.8°; Figs. 6A and 7D). Significantly, the ChRM mean direction of more distal granites (150 m from the dyke) is antipodal (D = 340.1°, I = -31.0°, k = 16.2, α_{95} = 17.1°; Figs. 6A and 7E). This may be explained in two ways. Firstly, there are two different granitic bodies with close but of slightly different in age, i.e. at least one geomagnetic reversal could occur after the emplacement of the older granite. Secondly, there could be a bipolar remagnetisation, but the age of this suggests reprint that should be older than the dyke age.

In any case, the contact test is positive, so that the dyke remanence is primary. The mean direction of the unbaked granites (Table 2, entry 6) probably primary and gives a Virtual Geomagnetic Pole (VGP), which is supported by the inverse contact test and by the presence of two polarities. Unfortunately, we have no date for this pole at present.

Despite these uncertainties, we conclude that the ChRM direction of the 1985 Ma intrusions (Table 2, entries 1 and 5) is primary.

5.2. Group of the 1975 Ma intrusions: the Deda Island dyke

The studied 1–2 m thick dyke is composed of fine- to medium-grained dolerite. It cuts across the ENE – trending 500 m thick Shalskiy gabbronorite dyke, which has been dated at ca 2504 Ma

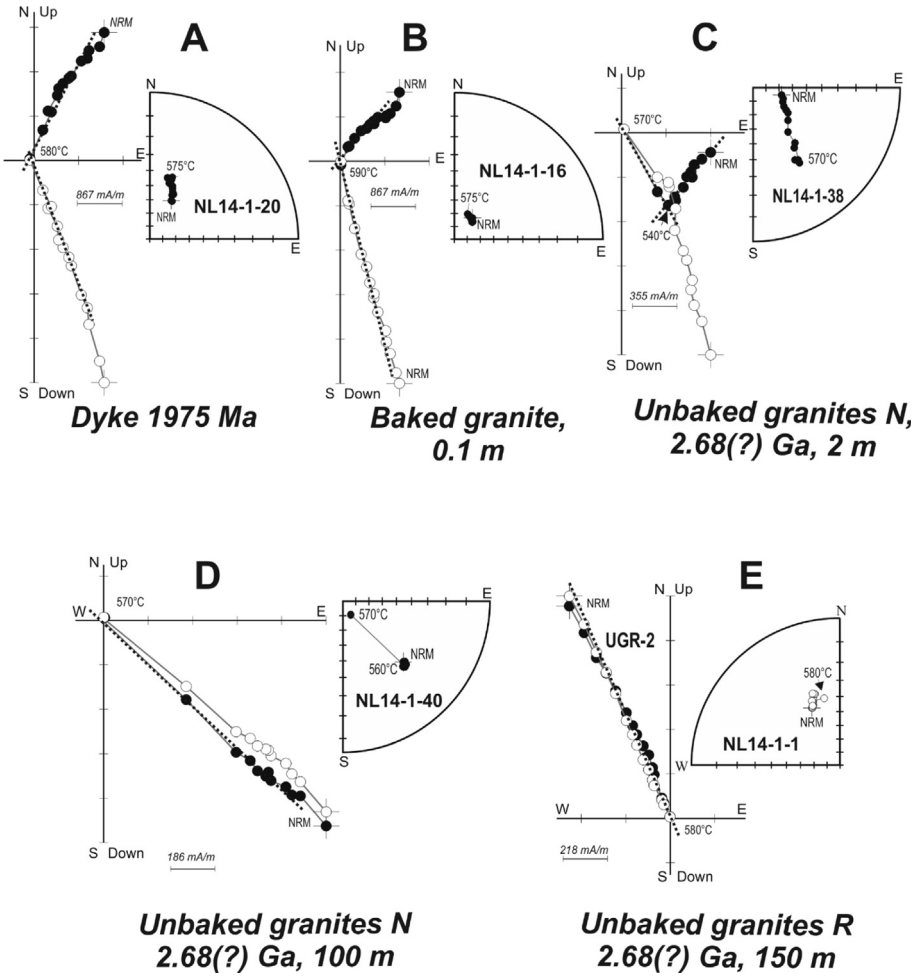


Fig. 7. Contact test of the Krestoviy Navolok dyke. Demagnetisations of: A – the dyke; B – the baked granite within 10 cm from the dyke; C – the granite at 2 m from the dyke; D – the granite 100 m from the dyke; E – the granite 150 m from the dyke.

(Bleeker et al., 2008). Both dykes also cut the Neoarchean granites (Fig. 5). We have studied the contacts of the 1975 Ma dyke both against the granites and the Shalskiy gabbronorite dyke. The demagnetisation behaviour of the Deda dyke (Table 2, entry 7; Fig. 8A) is typical for all the 1975 Ma intrusions (Fig. 3B–C). Baked samples of the 2504 Ma dyke, which have been collected within 10 cm from the contact with the 1975 Ma dyke, were entirely remagnetised by the younger dyke (Figs. 6B and 8B). Their ChRM directions are undistinguishable: D = 81.4°, I = 60.1°, k = 20.5, α_{95} = 5.7° in the 1975 Ma dyke and D = 91.0°, I = 61.4°, k = 23.1, α_{95} = 12.8° in the baked 2504 Ma gabbronorite. Samples collected within 30–70 cm from the contact with the younger dyke show the presence of two remanence components: less stable remanence with the direction close to the ChRM direction of the younger dyke and the high stable shallow SSW component (Fig. 8C–D). The stability of the former decreases with increasing distance from the younger dyke. The unbaked 2504 Ma gabbronorite dyke carries a stable one-component remanence (Table 2, entry 8; Figs 6B and 8E). This direction (D = 173.0°, I = -5.7°, k = 23.2, α_{95} = 6.3°) is close to the ChRM direction of this dyke published by Mertanen et al. (2006): D = 174.0°, I = 10.4°, k = 23.8, α_{95} = 16.0°. Hence, the ChRM of 1975 Ma Deda dyke is now supported by the inverse contact test and is likely to be primary. The mean ChRM direction of the baked gabbronorites of the 2504 Ma dyke (D = 91.0°, I = 61.4°, k = 23.1, α_{95} = 12.8°) is almost undistinguishable from the ChRM direction of the Deda dyke, but significantly different from the ChRM direction of the 2504 Ma gabbronorite dyke (Table 2, entry 8; Fig. 6B), and therefore, the contact test is positive.

The Neoarchean granites collected at the distance of 10 cm from the 1975 Ma dyke were not entirely remagnetised by the dyke, but reveal a low-stability component with the direction similar to that of the dyke (Fig. 8F). The unbaked granites collected at the distance of 250 m from the 1975 Ma dyke carry a consistent ChRM with steep upward SWW direction (Table 2, entry 9; Fig. 8G). This direction is significantly different from the ChRM directions of both 2504 and 1975 Ma dykes (Fig. 6B). We interpret this result as a tentatively positive contact test for the 1975 Ma dyke.

5.3. Group of the 1975 Ma intrusions: the Gorely Island dykes

Two thin (1–2 m) dykes on the Gorely Island (Fig. 5) are very similar in the appearance and composition with the nearby Deda Island dyke (see Section 5.2). We have collected baked granites near the contacts with both dykes as well as unbaked granites. In this case, all the granite samples within 10 cm from the dykes were entirely remagnetised by the 1975 Ma dykes (Fig. 6C). The mean ChRM direction of the baked granites (D = 75.7°, I = 62.9°, k = 37.4, α_{95} = 10.0°) is statistically undistinguishable from the ChRM directions of both Gorely Island dykes (D = 78.2°, I = 62.7°, k = 40.1, α_{95} = 10.7° and D = 82.0°, I = 62.1°, k = 35.6, α_{95} = 11.4°, Fig. 6C). The ChRM direction of unbaked granites is similar to that of the unbaked granites on the Deda Island (Fig. 6B and C). Despite

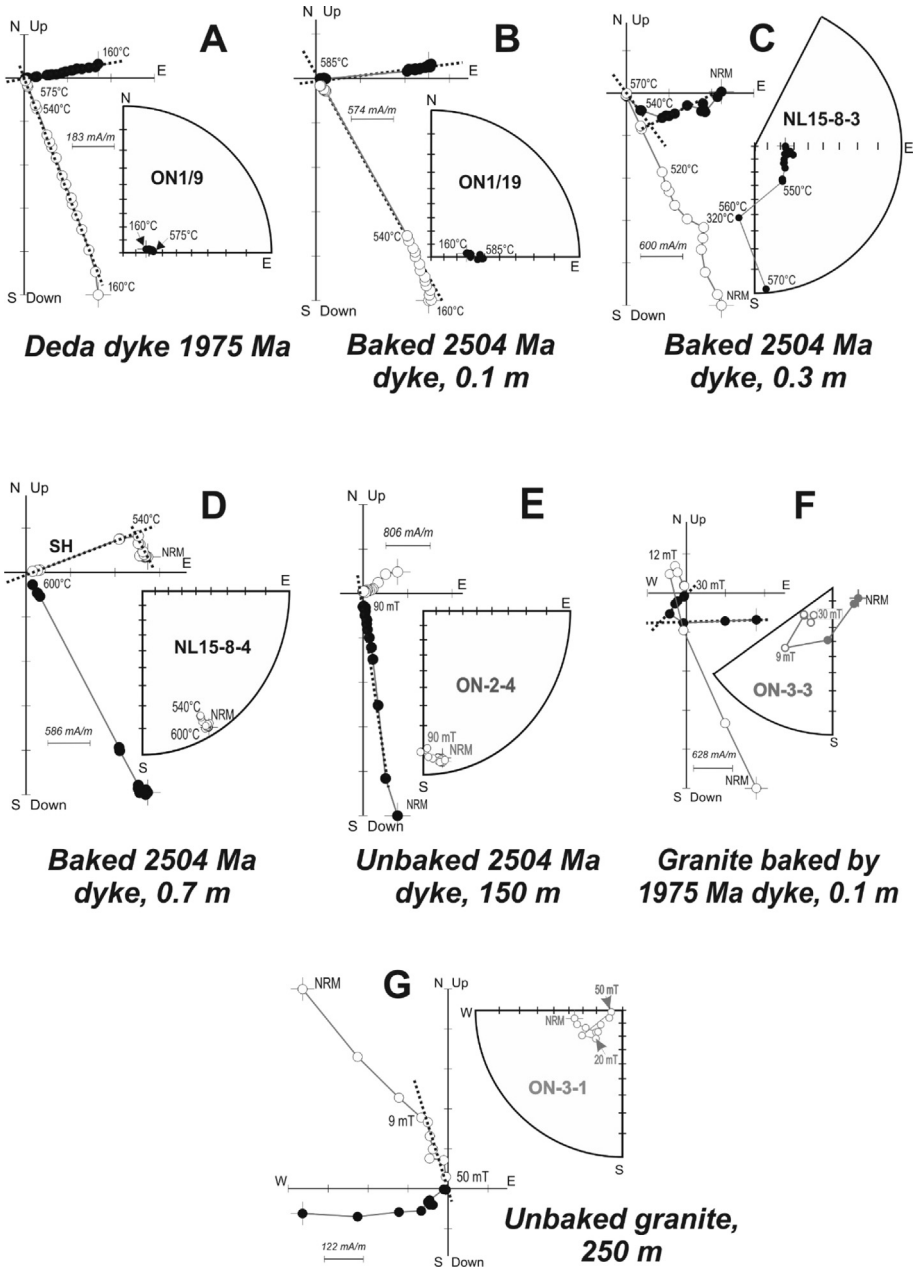


Fig. 8. Contact test of the Deda 1975 Ma dyke. Demagnetisations of: A – the dyke; B – the baked 2504 Ma gabbronorite dyke within 10 cm from the younger dyke; C – the 2504 Ma gabbronorite dyke within 30 cm from the younger dyke; D – the 2504 Ma gabbronorite dyke within 70 cm from the younger dyke; E – the unbaked 2504 Ma gabbronorite dyke 150 m from the younger dyke; F – the baked granite within 10 cm from the 1975 Ma dyke; G – the unbaked granite 250 m from the 1975 Ma dyke.

the absence of the hybrid zone with co-existing primary and overprinting remanences we interpret this contact test as positive.

5.4. Group of the 1975 Ma intrusions: the Shala dykes

We have studied three thin (1–2 m) 1975 Ma Shala dykes (Fig. 5) of the comparable composition to the Deda and Gorely dykes. Their contacts are not suitable for the paleomagnetic sampling, but we have collected instead eleven samples of the unbaked host granites. The ChRM directions of all these dykes are close to in the other 1975 Ma intrusions, and the host granites have ChRM directions similar to the granites on the Deda and Gorely islands (Fig. 6D). This is not a contact test, but it supports the positive contact tests of the Deda and Gorely dykes.

5.5. Contact test: summary

The positive contact tests imply that the ChRM directions of all the studied dykes are primary. Consequently, we conclude that: (i) the mean paleomagnetic direction and paleopole of the Pudozhgora intrusion and Krestoviy Navolok dyke can be at ca. 1985 Ma (Table 2, entries 1, 5 and Mean 1985 Ma); (ii) the mean paleomagnetic direction and paleopole of Gabnev sill, Gabnev dyke, Unoi sill, Deda and Gorely island dykes, Shala dykes and Suna River Canyon dolerites can be at ca. 1975 Ma (Table 2, entries 2–4, 7, 10–11, 13–15, 18 and Mean 1975 Ma).

Thus, the positive baked contact tests indicate that we have obtained two new Fennoscandian paleopoles, at ca. 1985 and ca. 1975 Ma (Table 2). The 1975 Ma one can be considered as a new Fennoscandian key pole: it satisfies 6 out of 7 points of the Q reli-

ability criteria (Van der Voo, 1990): it is well dated, with sufficient statistics, demagnetizations are adequate, positive contact tests, the area is tectonically coherent with craton and it does not resemble any younger reliable pole (Torsvik et al., 2012; Lubnina et al., 2014; Li et al., 2008; Pisarevsky et al., 2014). Our 1975 Ma pole also satisfies two basic criteria for key poles of Buchan (2014): it is clearly demonstrated to be primary and the rock units are precisely and accurately dated.

The 1985 Ma pole is also supported by a positive contact test, but it is based only on two intrusions, so the paleomagnetic secular variations are not fully averaged. This pole should be considered as a VGP and treated consciously in paleogeographic implications.

The paleomagnetic directions of the unbaked host granites for the Deda, Gorely and Shala dykes are stable and consistent between three sites (Table 2). Nesterova et al. (2011) reported the 2630 ± 10 Ma U–Pb (titanite) age for the Archean granites in the same area (Vodla River, village Kashino). We tentatively assign this age to the corresponding paleopole (Table 2, last entry).

6. Implications for the 1975 Ma paleogeography

The global paleogeography at 1975 Ma is poorly known. The participants of the 2014 Nordic Supercontinent Workshop in Haraldvalgen (Norway) made a new compilation of reliable Precambrian paleomagnetic poles (unpublished report). This compilation demonstrated that the paleomagnetic record for Fennoscandia is very deficient for the period between ca. 2400 and 1800 Ma. There are only two published paleopoles supported by field tests: one from the 2125 Ma Kuetsyarvi Formation in the Kola Peninsula (Torsvik and Meert, 1995) and another from the abovementioned 1975 Ma Konchozero sill. Both poles should be treated with caution: (i) Torsvik and Meert (1995) reported the positive conglomerate test, but they admitted that in many cases the intraformational boulders of the Kuetsyarvi Formation are much more stable than the parent rocks, and (ii) the Konchozero pole should be considered as VGP. Our new 1975 Ma pole is well dated and supported by robust contact tests, so that it can be considered as the key Fennoscandian pole.

As mentioned in the Introduction, numerous and various orogenic events commenced around 1975 Ma (e.g. Zhao et al., 2009). Many cratons (protocratons) were probably separated by oceans and started to amalgamate into larger continents such as Laurentia and Australia, and others, like Fennoscandia or Amazonia, and experienced accretionary growth (e.g. Hoffman, 1988; Bogdanova et al., 2008; Rosen et al., 1994; Pisarevsky et al., 2008; Cawood and Korsch, 2008; Hanson et al., 2011; Brito Neves, 2011). However, there are only three reliable paleomagnetic poles from different continents with ages close to 1975 Ma (Table 3). The Western River Formation paleopole from the Slave Craton reported by Evans and Hoye (1981) and was supported by the inverse contact test. Bowring and Grotzinger (1992) dated this formation at ca 1963 Ma (U–Pb, zircon). By that time, the stratigraphic nomenclature in northern Slave was revised, and the Red Siltstone member of the Western River Formation was renamed as to the Rifle Formation (Grotzinger et al., 1987). This pole is included into Table 2 under the new name. The 1998 Ma Minto dykes pole of Buchan et al. (1998) from the Western Superior Craton was recalculated by Evans and Halls (2010). One of the Minto dykes cuts across the older Ptarmigan dyke, but its contacts are not exposed. The remanence directions in these two dykes are different, which support the primary remanence of the Minto dyke (Buchan et al., 1998). However, this is not a robust contact test. Bispo-Santos et al. (2014) reported the 1980–1960 Ma Surumu Volcanics Amazonian pole. This pole is supported by inverse contact test.

Using our 1975 Ma robust Karelian pole, we have now four reliable and approximately coeval paleopoles for Karelia, Western Superior, Slave and Amazonia. This provides an opportunity to restore paleolatitudinal positions and azimuthal orientations of these cratons (Fig. 9A). Even though this is not an unequivocal paleogeographic reconstruction in view of polarity ambiguity and longitudinal uncertainty, we consider this one as preferable, also adding some other cratons by the following reasons.

1. Some recent pre-2.0 Ga reconstructions (e.g. Bleeker and Ernst, 2006; Söderlund et al., 2010; Pisarevsky et al., 2015) suggest a connection between Karelia and Superior. However, the mutual position of these cratons in the paleomagnetically and geologically supported 1770 Ma to 1270 Ma reconstructions (e.g. Stearn and Piper, 1984; Gower et al., 1990; Park, 1992; Pisarevsky and Bylund, 2010) is different – they are separated by the Nain and Rae cratons and two 1.9–1.8 Ga orogens (Torngat and Nagssugtoqidian – see Fig. 9C). This implies that Karelia and Superior were separated at 1.98 Ga. In our 1975 Ma paleomagnetically permissive reconstruction (Fig. 9A) Nain and Rae cratons (shapes as in Hoffman, 1988) are located between Fennoscandia and Superior. Thus, Karelia should separate from the Superior probably before 2.0 Ga, then moved independently and docked to the assembled Laurentia by c. 1.78 Ga. This is generally consistent with inferred breakup of Karelia from southern Superior at c. 2070–2000 Ma (e.g. Bleeker and Ernst, 2006). We locate Karelia in the 1975 Ma reconstruction in accord to our paleomagnetic data, but on a moderate distance from Superior giving a space for the Rae and Nain cratons to join Superior in Laurentia before the docking of Fennoscandia, and a space for the accretionary growth of Fennoscandia during Svecofennian orogeny (Bogdanova et al., 2015).
2. Proposing the paleogeographic model of the Laurentian assembly, Whitmeyer and Karlstrom (2007) suggested only the small "Thelon" ocean located between Slave and Rae cratons at ca. 2000 Ma. They have supposed that this ocean has been closed at 1960–1920 Ma by docking of the Slave Craton to the Rae Craton. They have also suggested that by that time Rae and Hearn cratons were already united and separated from Superior also by a relatively small "trans-Hudson" ocean, which was closed by 1820–1800 Ma by the amalgamation of the composite Slave/Rae/Hearne with Superior. However, the paleomagnetic data in Table 3 require a long latitudinal distance and large azimuthal rotation between Slave and Superior at ca. 1975 Ma. In contrast to the model of Whitmeyer and Karlstrom (2007), Berman et al. (2007) have hinted that Rae and Hearne were jointed together only at ca. 1900 Ma along the Snowbird tectonic zone. In our reconstruction (Fig. 9A, B) we attempt to find a compromise between the 1.98 Ga paleomagnetic data and these models. Consequently, we placed the Slave, Rae and Hearne cratons as close as possible to Superior, but in accord to the paleomagnetic data (Fig. 9B and C).
3. The longitudinal position of Amazonia is also unconstrained. The popular SAMBA model (Johansson, 2009) in our view contradicts geological and paleomagnetic data (Pisarevsky et al., 2014; Bogdanova et al., 2015; Bogdanova and Pisarevsky, 2015). For this reason, we place Amazonia at somewhat paleomagnetically permitted distance from Karelia/Fennoscandia.

In our reconstructions we have adopted the model of Evans and Halls (2010) of the post-2070 Ma clockwise rotation of Eastern Superior with respect to Western Superior around the Euler's pole of 51°N, 85°W of 14° (Table 4), suggesting that the rotation did not yet occur at 1975 Ma (Fig. 9). Rotation parameters for the 1775 Ma reconstruction are as in Pisarevsky et al. (2014).

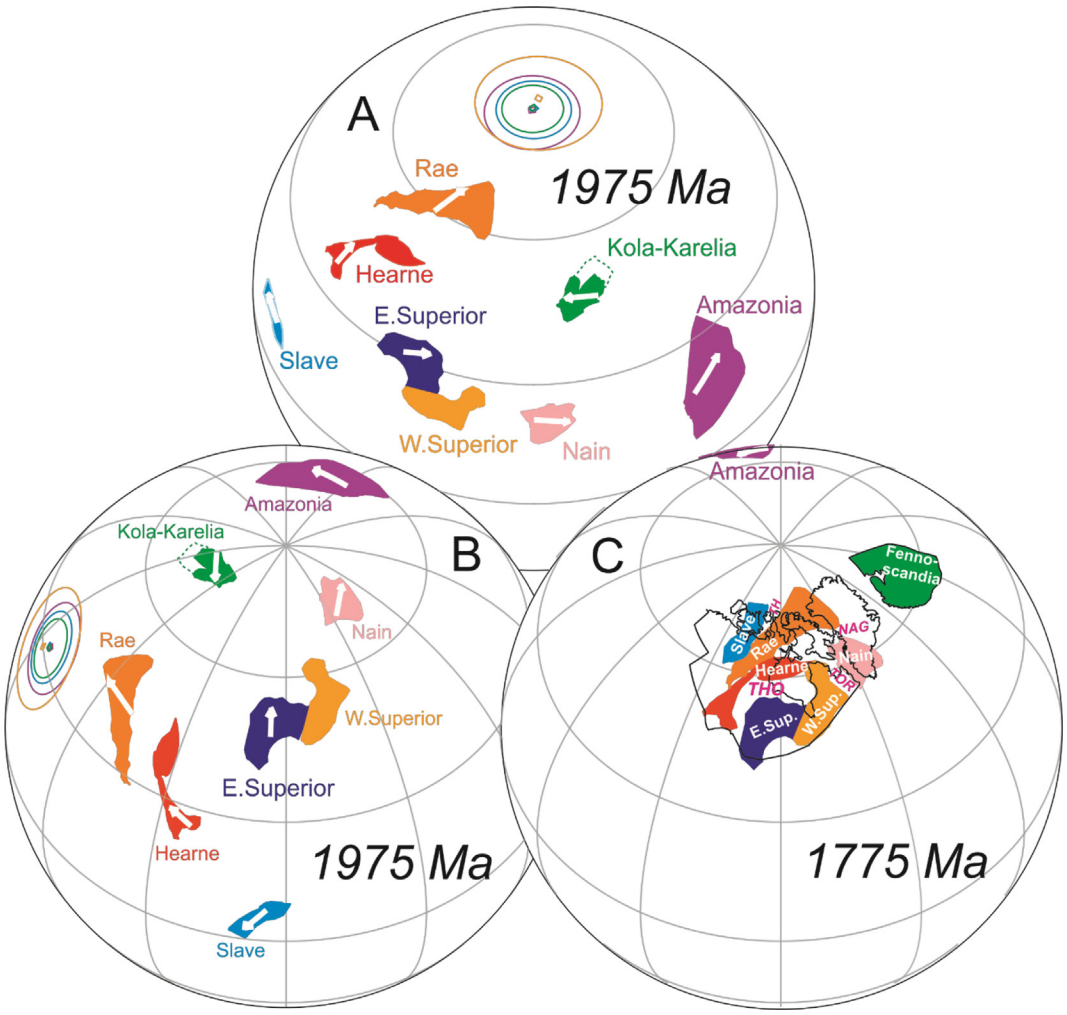


Fig. 9. Paleogeographic reconstructions: A – 1975 Ma, absolute coordinates; B – 1975 Ma, East Superior Craton coordinates; C – 1775 Ma, Laurentian coordinates. Paleopoles are from Table 3, colour code is the same as for cratons. 1.9–1.8 Ga orogens: THO – trans-Hudson; TOR – Torngat; NAG – Nagssugtoqidian; TH – Thelon. Arrows show direction to the North.

Table 4
Euler rotation parameters for Fig. 9A (1975 Ma reconstruction).

Craton/block/terrane	Pole		Angle
	(°N)	(°E)	(°)
	1975 Ma		
W. Superior to abs. framework	48.47	−123.24	−93.58
E. Superior to W. Superior	51.00	−85.00	14.00
Rae to W. Superior	−23.96	89.73	−55.14
Hearne to Rae	52.40	−42.78	14.18
Nain to W. Superior	57.85	−20.15	−40.76
Slave to W. Superior	38.13	−90.29	156.47
Karelia to W. Superior	60.27	−58.24	67.23
Amazon to W. Superior	0.68	−35.68	−61.98
	1775 Ma		
Fennoscandia to Laurentia	44.99	7.45	44.93
Greenland to Laurentia	67.50	−118.48	−14.00

The 1975 Ma reconstruction (Fig. 9A, B) is not unique, but it is paleomagnetically permissible and not unreasonable geologically. However, two paleogeographically important features should be noted: (i) Karelia, Superior and Slave cratons were located at significantly different paleolatitudes, and (ii) their mutual azimuthal orientations require significant rotations before their final assembly at ca. 1775 Ma (Fig. 9C). This implies that the amalgamation of building blocks of the Paleo-Mesoproterozoic supercontinent Nuna/Columbia probably started after 1975 Ma.

7. Conclusions

1. The paleomagnetic study of the 1985–1975 Ma mafic dykes and sills in southern Karelia, Russia has revealed the presence of stable primary remanence, supported by robust contact tests and rock magnetic analysis.
2. The new 1975 Ma paleomagnetic pole can be qualified as the key pole for Fennoscandia.
3. The 1985 Ma remanence of two intrusions, maintained by the robust contact test, is also primary, and the corresponding paleomagnetic pole is considered as well dated VGP.
4. The paleopole of the host Archean granites in the Vodlozero terrane (the Shala area) with a tentative ca. 2630 Ma age is supported by inverse contact test. We provide an additional analysis and a positive inverse contact test to the previously paleomagnetically studied 2504 Ma Shalskiy gabbronorite dyke.
5. We propose a paleomagnetically allowable 1975 Ma paleogeographic reconstruction of Karelia, Superior, Slave, Rae, Hearne and Amazonia cratons. Their paleopositions suggest that the

process of amalgamation of building blocks of the Paleo-Mesoproterozoic supercontinent Nuna/Columbia started only after 1975 Ma.

Acknowledgements

We are indebted to Alexander Slabunov and Alexander Pasenko for their participation in the fieldwork, technical support for fieldwork and sample preparation. Reviews by two anonymous reviewers are appreciated, as is the editorial handling by Prof. Guochun Zhao.

This work was supported by Russian Fund of Basic Research, grant #14-05-00731. All measurements were carried out on the equipment purchased by Development Programme of the M.V. Lomonosov Moscow State University (ID 10932885). The geochemical studies were supported by RSF grant 16-17-10260. This is contribution 882 from the ARC Centre of Excellence for Core to Crust Fluid Systems (http://www.ccfs.mq.edu.au) and contribution to IGCP 648 (http://geodynamics.curtin.edu.au/igcp-648/). SAP's work was supported by Australian Research Council through Australian Laureate Fellowship grant to Z.-X. Li.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2017.01.011.

References

Amelin, Y.V., Heaman, L.M., Semenov, V.S., 1995. U-Pb geochronology of layered mafic intrusions in the eastern Baltic Shield: implications for the timing and duration of Paleoproterozoic continental rifting. Precambr. Res. 75, 31–46.

Berman, R.G., Davis, W.J., Pehrsson, S., 2007. Collisional Snowbird tectonic zone resurrected growth of Laurentia during the 1.9 Ga accretionary phase of the Hudsonian orogeny. Geology 35, 911–914.

Bispo-Santos, F., D'Agrella-Filho, M.S., Janikian, L., Reis, N.J., Trindade, R.I.F., Reis, M. A.A.A., 2014. Towards Columbia: Paleomagnetism of 1980–1960 Ma Surumuvolcanic rocks, Northern Amazonian Craton. Precambr. Res. 244, 123–138.

Bleeker, W., Ernst, R.E., 2006. Short-lived mantle generated magmatic events and their dike swarms: the key unlocking Earth's paleogeographic record back to 2.6 Ga. In: Hanski, E., Mertanen, S., Rämö, T., Vuollo, J. (Eds.), Dike Swarms: Time Markers of Crustal Evolution. Taylor & Francis/Balkema, Leiden, pp. 3–26.

Bleeker, W., Hamilton, M.A., Ernst, R.E., Kulikov, V.S., 2008. The search for Archean-Paleoproterozoic supercratons: new constraints on Superior-Karelia-Kola correlations within supercraton Superia, including the first ca. 2504 Ma (Mistassini) ages from Karelia, in: Abstract for IGC33 Meeting, Oslo, Norway.

Bogdanova, S.V., Gorbatschev, R., Skridlaite, G., Soesoo, A., Taran, L., Kurlovich, D., 2015. Trans-Baltic Palaeoproterozoic correlations towards there construction of supercontinent Columbia/Nuna. Precambr. Res. 259, 5–33.

Bogdanova, S.V., Gintov, O.B., Kurlovich, D.M., Lubnina, N.V., Nilsson, M.K.M., Orlyuk, M.I., Pashkevich, I.K., Shumlyanskyy, L.V., Starostenko, V.I., 2013. Late Palaeoproterozoic mafic dyking in the Ukrainian Shield of Volgo-Sarmatia caused by rotation during the assembly of supercontinent Columbia (Nuna). Lithos 174, 196–216.

Bogdanova, S.V., Bingen, B., Gorbatschev, R., Kheraskova, T.N., Kozlov, V.I., Puchkov, V.N., Volozh, Yu.A., 2008. The East European Craton (Baltica) before and during the assembly of Rodinia. Precambr. Res. 160, 23–45.

Bogdanova, S.V., Pisarevsky, S.A., 2015. Were Amazonia and Baltica Connected in Nuna and Rodinia? AGU Fall Meeting, San Francisco, USA. T13A-2966.

Bowring, S.L., Grotzinger, J.P., 1992. Implications of new chronostratigraphy for tectonic evolution of Wopmay orogeny, northwest Canadian shield/. Am. J. Sci. 292, 1–20.

Brito Neves, B.B., 2011. The Paleoproterozoic in the South-American continent: diversity in the geologic time. J. S. Am. Earth Sci. 32, 270–286.

Buchan, K.L., 2014. Key paleomagnetic poles and their use in Proterozoiccontinent and supercontinent reconstructions: a review. Precambr. Res. 244, 5–22.

Buchan, K.L., Mortensen, J.K., Card, K.D., Percival, J.A., 1998. Paleomagnetism and U-Pb geochronology of diabase dyke swarms of Minto block, Superior Province, Quebec. Canada. Can. J. Earth Sci. 35, 1054–1069.

Buchan, K.L., Mitchell, R.N., Bleeker, W., Hamilton, N.A., LeCheminant, A.N., 2016. Paleo- magnetism of ca. 2.13–2.11 Ga Indin and ca. 1.885 Ga Ghost dyke swarms of the Slave craton: Implications for the Slave craton APW path and relative drift of Slave, Superior and Siberian cratons in the Paleoproterozoic. Precambrian Res. 275, 151–175.

Cawood, P.A., Korsch, R.J., 2008. Assembling Australia: Proterozoic building of a continent. Precambrian Res. 166, 1–38.

Daly, J.S., Balagansky, V.V., Timmerman, M.J., Whitehouse, M.J., 2006. The Lapland-Kola orogen: Palaeoproterozoic collision and accretion of the northern Fennoscandian lithosphere. Geological Society, London, Memoirs 32, 579–598.

Ernst, R.E., 2014. Large Igneous Provinces. Cambridge University Press, p. 653.

Evans, D.A.D., Halls, H.C., 2010. Restoring Proterozoic deformation within the Superior craton. Precambrian Res. 183, 474–489.

Evans, M.E., Hoye, G.S., 1981. Paleomagnetic results from the Lower Proterozoic rocks of Great Slave Lake and Bathurst areas, Northwest Territories. Geological Survey of Canada, Paper 81–10, 191–202.

Glushanin, L.V., Sharov, N.V., Shiptsov, V.V. (Eds.), 2011. Paleoproterozoic Onega Structure (Geology, Tectonics, Deep Structure and Mineralogeny). Karelian Sci. Center RAS, Petrozavodsk, p. 431.

Golubev, A.I., Svetov, A.P., 1983. Geochemistry of Platform Basalts. Karelia, Petrozavodsk (In Russian), p. 190.

Gower, C.F., Ryan, A.F., Rivers, T., 1990. Mid-Proterozoic Laurentia–Baltica: an overview of its geological evolution and a summary of the contributions made by this volume. In: Gower, C.F., Rivers, T., Ryan, B. (Eds.), Mid-Proterozoic Laurentia-Baltica. Geological Association of Canada, St. John's Newfounland, pp. 23–40.

Grotzinger, J.P., McCormick, D.S., Pelechaty, S.M., 1987. Progress report on the stratigraphy, sedimentology and significance of the Kimerot and Bear Creek groups, Kilohigok Basin, District of Mackenzie. In Current Research, Part A, Geological Survey of Canada Paper 87–1A, 219–238.

Hanson, R.E., Rioux, M., Gose, W.A., Blackburn, T.J., Bowring, S.A., Mukwakwami, J., Jones, D.L., 2011. Paleomagnetic and geochronological evidence for large-scale post-1.88 Ga displacement between the Zimbabwe and Kaapvaal cratons along the Limpopo belt. Geology 39, 487–490.

Hoffman, P.F., 1988. United plates of America, the birth of a craton: early Proterozoic assembly and growth of Laurentia. Annu. Rev. Earth Planet. Sci. 16, 543–603.

Johansson, Å., 2009. Baltica, Amazonia and the SAMBA connection—1000 million years of neighbourhood during the Proterozoic? Precambrian Res. 175, 221–234.

Kirschvink, J.L., 1980. The least−squares line and plane and the analysis of paleomagnetic data. Geophys. J. Royal Astron. Soc. 62, 699–718.

Kulikov, V.S., Bychkova, Ya, V., Kulikova, V.V., Ernst, R., 2010. The Vetreny Poyas (Windy Belt) subprovince of southeastern Fennoscandia: an essential component of the ca. 2.5–2.4 Ga Sumian large igneous provinces. Precambrian Res. 183, 589–601.

Kulikov, V.S., Kulikova, V.V., 2008. Geology of the Precambrian basement in the territory of the Kivach National Park. Geologia I poleznye iskopaemye Karelii 11, 103–109 (In Russian).

Kulikov, V.S., Kulikova, V.V., Lavrov, B.S., Puchtel, I.S., Pisarevsky, S.A., Sokolov, S.J., 1999. Palaeoproterozoic Suisarian picrit-basaltic Complex in Karelia (Basic Section and Petrology). Karelian Research Centre of Russian Academy of Science, Petrozavodsk, p. 96 (in Russian).

Kulikov, V.S., Svetov, S.A., Slabunov, A.I., Kulikova, V.V., Polin, A.K., Golubev, A.I., Gorkovets, V.Y., Ivashchenko, V.I., Gogolev, M.A., 2016. Geological map of southeaster Fennoscandia (scale 1:750 000): a new approach to map compilation. Proc. Karelian Res. Centre RAS, 1–41 (in Russian).

Li, Z.X., Bogdanova, S.V., Collins, A., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I., Fuck, R., Gladkochub, D., Jacobs, J., Karlstrom, K., Lu, S., Milesi, J.-P., Myers, J., Natapov, L., Pandit, M., Pease, V., Pisarevsky, S.A., Thrane, K., Vernikovsky, V., 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Res. 160, 179–210.

Lubnina, N.V., Stepanova, A.V., Ernst, R.E., Nilsson, M., Söderlund, U., 2016. New U-Pb baddeleyite age, and AMS and paleomagnetic data for dolerites in the Lake Onega region belonging to the 1.98–1.95 Ga regional Pechenga-Onega Large Igneous Province. GFF 138, 54–78.

Lubnina, N.V., Pasenko, A.M., Novikova, M.A., Bubnov, A.Y., 2015a. The East European craton at the end of the Paleoproterozoic: a new paleomagnetic pole of 1.79–1.75 Ga. Moscow Univ. Geol. Bull. 70 (6), 495–503.

Lubnina, N.V., Zakharov, V.S., Novikova, M.A., Vorontsova, V.P., 2015b. Paleoproterozoic remagnetization in the White Sea Mobile Belt, Karelia: Petro-Paleomagnetic evidence and supercomputer modeling. Moscow Univ. Geol. Bull. 70 (2), 84–96. http://dx.doi.org/10.3103/S0145875215020052.

Lubnina, N.V., Pisarevsky, S.A., Puchkov, V.N., Kozlov, V.I., Sergeeva, N.D., 2014. New paleomagnetic data from Late Neoproterozoic sedimentary successions in Southern Urals, Russia: implications for the Late Neoproterozoic paleogeography of the Iapetan realm. Int. J. Earth Sci. 103, 317–334.

Malashin, M.V., Golubev, A.I., Ivanikov, V.V., Philippov, N.B., 2003. Geochemistry and petrology of Paleoproterozoic Mafic Volcanic Complexes of Karelia I: Jatuli Volcanic Complex. S.-Petersb. State Univ. Bull., Series Geol., Geogr. 1, 3–19.

Martin, A.P., Prave, A.R., Condon, D.J., Lepland, A., Fallick, A.E., Romashkin, A.E., Medvedev, P.V., Rychanchik, D.V., 2015. Multiple Palaeoproterozoic carbon burial episodes and excursions. Earth Planet. Sci. Lett. 424, 226–236.

Meert, J.G., 2012. What's in a name? The Columbia (Paleopangaea/Nuna) supercontinent. Gondwana Res. 21, 987–993.

Mertanen, S., Vuollo, J.I., Huhma, H., Arestova, N.A., Kovalenko, A., 2006. Early Paleoproterozoic-Archean dykes and gneisses in Russian Karelia of the Fennoscandian Shield – New paleomagnetic, isotope age and geochemical investigations. Precambrian Res. 144, 239–260.

Melezhik, V.A., Hanski, E.J., 2013. Palaeotectonic and palaeogeographic evolution of Fennoscandia in the early Palaeoproterozoic. In: Melezhik, V.A., Prave, A.R., Fallick, A.E., Kump, L.R., Strauss, H., Lepland, A., Hanski, E.J. (Eds.), Reading the Archive of Earth's Oxygenation, Volume 1: The Palaeoproterozoic of Fennoscandia as Context for the Fennoscandian Arctic Russia – Drilling Early Earth Project, Frontiers in Earth Sciences. Springer, Berlin, Heidelberg, pp. 111–178.

Melezhik, V.A., Medvedev, P.V., Svetov, S.A., 2013. The Onega basin. In: Melezhik, V. A., Prave, A.R., Fallick, A.E., Kump, L.R., Strauss, H., Lepland, A., Hanski, E.J. (Eds.), Reading the Archive of Earth's Oxygenation, Volume 1: The Palaeoproterozoic of Fennoscandia as Context for the Fennoscandian Arctic Russia – Drilling Early Earth Project, Frontiers in Earth Sciences. Springer, Berlin, Heidelberg, pp. 387–490.

Nesterova, N.S., Kirnozova, T.I., Fugzan, M.M., 2011. U-Pb titanite ages of the rocks of Karelian Craton and Belomorian mobile belt of the Fennoscandian shield (new data). Geokhimiya N12, 1235–1242 (in Russian).

Park, R.G., 1992. Plate kinematic history of Baltica during the Middle to Late Proterozoic: a model. Geology 20, 725–728.

Pasenko, A.M., Lubnina, N.V., 2014. The Karelian Craton in the Paleoproterozoic: new paleomagnetic data. Moscow Univ. Geol. Bull. 69, 189–197.

Philippov, N.B., Trofimov, N.N., Golubev, A.I., Sergeev, S.A., Huhma, H., 2007. New geochronological data on Koikarsko-Svyatnavoloksky and Pudozhgorsky gabbro-dolerite intrusions. Geol. Mineral Resour. Karelia 10, 49–68 (in Russian).

Pisarevsky, S.A., Bylund, G., 2010. Paleomagnetism of 1780–1770 Ma mafic and composite intrusions of Småland (Sweden): implications for the Mesoproterozoic supercontinent. Am. J. Sci. 310, 1168–1186.

Pisarevsky, S.A., Sokolov, S.J., 1999. Palaeomagnetism of the Palaeoproterozoic ultramafic intrusion near Konchozero Lake, Southern Karelia, Russia. Precambrian Res. 93, 201–213.

Pisarevsky, S.A., De Waele, B., Jones, S., Söderlund, U., Ernst, R.E., 2015. Paleomagnetism and U-Pb age of the 2.4 Ga Erayinia mafic dykes in the south-western Yilgarn, Western Australia: Paleogeographic and geodynamic implications. Precambrian Res. 259, 222–231.

Pisarevsky, S.A., Elming, S.-Å., Pesonen, L.J., Li, Z.X., 2014. Mesoproterozoic paleogeography: supercontinent and beyond. Precambrian Res. 244, 207–225.

Pisarevsky, S.A., Natapov, L.M., Donskaya, T.V., Gladkochub, D.P., Vernikovsky, V.A., 2008. Proterozoic Siberia: a promontory of Rodinia. Precambrian Res. 160, 66–76.

Puchtel, I.S., Arndt, N.T., Hofmann, A.W., Haase, K.M., Kröner, A., Kulikov, V.S., Kulikova, V.V., Garbe-Schönberg, C.-D., Nemchin, A.A., 1998. Petrology of mafic lavas within the Onega plateau, central Karelia: evidence for 2.0 Ga plume-related continental crustal growth in the Baltic Shield. Contrib. Mineral. Petrol. 130, 134–153.

Puchtel, I.S., Haase, K.N., Hofman, A.W., et al., 1997. Petrology and geochemistry of crustally contaminated komatiitic basalts from Vetreny Belt, southeastern Baltic Shield: evidence for an early Proterosoic mantle plume beneath rifted Archean continental lithosphere // Geochim. Cosmochim. Acta 61, 1205–1222.

Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic supercontinent. Gondwana Res. 5 (1), 5–22.

Rosen, O.M., Condie, K.C., Natapov, L.M., Nozhkin, A.D., 1994. Archean and Early Proterozoic evolution of the Siberian craton: a preliminary assessment. In: Condie, K.C. (Ed.), Archean Crustal Evolution. Elsevier, Amsterdam, pp. 411–459.

Salminen, J., Halls, H.C., Mertanen, S., Pesonen, L.J., Vuollo, J., Söderlund, U., 2014. Paleomagnetic and geochronological studies on Paleoproterozoic diabase dykes of Karelia, East Finland—Key for testing the Superia supercraton. Precambrian Res. 244, 87–99.

Slabunov, A.I., Lobach-Zhuchenko, S.B., Bibikova, E.V., Sorjonen-Ward, P., Balagansky, V.V., Volodichev, O.I., Shchipansky, A.A., Svetov, S.A., Chekulaev, V.P., Arestova, N.A., Stepanov, V.S., 2006. The Archaean nucleus of the Fennoscandian (baltic) shield. In: Gee, D.G., Stephenson, R.A. (Eds.), European Lithosphere Dynamics. Geological Society, London Memoirs 32, pp. 627–644.

Stearn, J.E.F., Piper, J.D.A., 1984. Palaeomagnetism of the Sveconorwegian mobile belt of the Fennoscandian Shield. Precambrian Res. 23, 201–246.

Stepanova, A.V., Salnikova, E.B., Samsonov, A.V., Egorova, S.V., Larionova, Yu.O., Stepanov, V.S., 2015. The 2.31 Ga mafic dykes in the Karelian craton, eastern Fennoscandian shield: U-Pb age, source characteristics and implications for continental break-up processes. Precambrian Res. 259, 43–57.

Stepanova, A.V., Samsonov, A.V., Larionov, A.N., 2014. The final episode of the Mid-Paleoproterozoic magmatism in the Onega Basin: data on dolerites in Zaonezhski Peninsula. Proc. Karelian Res. Centre RAS, 3–16.

Söderlund, U., Hofmann, A., Klausen, M.B., Olsson, J.R., Ernst, R.E., Persson, P.-O., 2010. Towards a complete magmatic barcode for the Zimbabwe craton: Baddeleyite U-Pb dating of regional dolerite dyke swarms and sill complexes. Precambrian Res. 183, 388–398.

Torsvik, T.H., Meert, J.G., 1995. Early Proterozoic palaeomagnetic data from the Pechenga zone (north-west Russia) and their bearing on Early Proterozoic palaeogeography. Geophy. J. Int. 122, 520–536.

Torsvik, T.H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P.V., van Hinsbergen, D.J.J., Domeier, M., Gaina, C., Tohver, E., Meert, J.G., McCausland, P.J.A., Cocks, R.M., 2012. Phanerozoic polar wander, palaeogeography and dynamics. Earth-Sci. Rev. 114, 325–368.

Trofimov, N.N., Golubev, A.I., 2008. Pudozhgorsky Noble Metal Titanomagnetite Ore Deposit. Karelian Research Centre RAS, Petrozavodsk, p. 123.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.

Whitmeyer, S.J., Karlstrom, K.E., 2007. Tectonic model for the Proterozoic growth of North America. GSA Bull. 3, 220–259.

Zhang, S., Li, Z.X., Evans, D.A.D., Wu, H., Li, H., Dong, J., 2012. Pre-Rodinia supercontinent Nuna shaping up: a global synthesis with new paleomagnetic results from North China. Earth Planet. Sci. Lett. 353–354, 145–155.

Zhao, G., He, Y., Sun, M., 2009. The Xiong'er volcanic belt at the southern margin of the North China Craton: Petrographic and geochemical evidence for its outboard position in the Paleo-Mesoproterozoic Columbia Supercontinent. Gondwana Res. 16, 170–181.

Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth Sci. Rev. 67, 91–123.

Zhao, G., Sun, M., Wilde, S.A., Li, S., Zhang, J., 2006. Some key issues in reconstructions of Proterozoic supercontinents. J. Asian Earth Sci. 28, 3–19.